

Incident Management System v1.0

Beta Test Checklist

Test Session Record

Tester (name / callsign):		Date:	
Device / browser used:		Server IP confirmed:	192.168.50.1
Overall result (circle):	PASS PASS WITH NOTES FAIL		

How to use this checklist

Work through each section in order. For each item, tick **PASS** if it works correctly, or **FAIL** and write a note if it does not. The Notes column is for brief observations — use extra paper for detail. Items marked with a hint in the Notes column are things to look for specifically. Submit any failures with the page name and what you expected vs. what happened.

PASS — Feature works as expected.

FAIL — Feature does not work, is missing, or produces wrong output.

1 · System & Infrastructure

Access, services, Wi-Fi AP, and health monitoring

1.1 Access & Network

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Dashboard loads at http://192.168.50.1 on a device connected to EMCOMM-NET	Try phone, tablet, laptop
☐	☐	SSID "EMCOMM-NET" visible and password accepted	WPA2
☐	☐	Dashboard loads at http://fieldcomms.local (mDNS alias)	
☐	☐	Dashboard loads at http://emcomm.local (mDNS alias)	
☐	☐	Single router coverage verified — EMCOMM-NET reachable throughout the deployment area	
☐	☐	All dashboard cards load without 404 errors	Click each card
☐	☐	UTC clock in nav bar updates every second	

1.1b AiMesh Coverage Extension (if deployed)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	If AiMesh nodes are deployed: all nodes paired and showing Online in AiMesh menu	Optional — only test if nodes are in use
☐	☐	EMCOMM-NET visible and http://192.168.50.1 reachable from each node coverage zone	Walk to each zone with a phone
☐	☐	Device roams between nodes automatically when moving across the venue	No manual Wi-Fi reconnection needed
☐	☐	Pi remains at 192.168.50.1 throughout — not affected by which node device connects through	
☐	☐	Wired backhaul nodes (if used): CAT 6 link confirmed by link LED on node LAN port	

1.2 Pre-Flight Health Check (<http://192.168.50.1/preflight.html>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Page loads and shows all 7 sections	Power, Radio, Winlink, Pi, Servers, Personnel, Logistics
☐	☐	Each checklist item can be set GO / CAUTION / NO-GO	Click the buttons
☐	☐	Overall status banner (GO / CAUTION / NO-GO) updates as items change	
☐	☐	Save State persists items after page reload	
☐	☐	Reset All clears all items back to unset	
☐	☐	Export Report downloads a PDF summary	
☐	☐	Print produces a clean printable page	

1.3 Dashboard Health Monitor (port 5051)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Dashboard health panel shows CPU temperature	
☐	☐	Dashboard health panel shows memory usage	
☐	☐	Dashboard health panel shows disk usage	
☐	☐	Service status shown for: nginx, FCC Lookup API, Health Monitor	
☐	☐	Service status shown for: Graywolf APRS, Pat Winlink, Kiwix	
☐	☐	Service status shown for: GPS Daemon, NTP/Chrony, Dead Man's Switch, ICS Platform	
☐	☐	Internet connectivity indicator shown	
☐	☐	GPS status shown (fix / no fix)	Requires USB GPS
☐	☐	Tailscale status shown	
☐	☐	Health data refreshes automatically	

1.4 Dashboard Mode Switcher

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Dashboard has Amateur Radio and ICS / Incident Command mode tabs	
☐	☐	Switching modes changes the visible card set	
☐	☐	Mode selection persists after page reload	
☐	☐	Three mode buttons visible in the mode bar: ■ Amateur Radio · ■ Starcom / Public Safety · ■ ICS	
☐	☐	Mode selection persists across browser sessions (saved to localStorage)	
☐	☐	Amateur mode — Net Ops: Net Control Logger, Observer Mode, JS8Call, APRS Tactical Map	
☐	☐	Amateur mode — Reference: Callsign Lookup, HF Propagation, Repeaters, Grid Square, Cheat Sheets	
☐	☐	Amateur mode — Forms: NTS Radiogram, ICS-213, ICS-214, Print Center, Reference Library	
☐	☐	Amateur mode — Safety: Dead Man's Switch, Roster, Pre-Flight Checklist	
☐	☐	Starcom mode — Net Ops: Starcom Net Logger, Weather Net, SAR Net, Observer Mode	Starcom is now its own dedicated mode
☐	☐	Starcom mode — Resources: Resource Tracking Map, Resource Board, Member Roster, Facilities	
☐	☐	Starcom mode — ICS: ICS Platform, ICS-213, ICS-214, Print Center	
☐	☐	ICS mode — Platform: ICS Command, Operations, Planning, Logistics, Finance, Planning P	
☐	☐	ICS mode — Resources: Tactical Map, Resource Map, Resource Board, Roster, Facilities	
☐	☐	ICS mode — Forms: ICS-213, ICS-214, ICS-309, Print Center, Reference Library	
☐	☐	ICS mode — Comms: Net Control, Starcom, Winlink Form Import, Repeater Database, Cheat Sheets	
☐	☐	URL param ?mode=starcom loads dashboard in Starcom mode directly	

2 · Net Control Logging

Amateur net logging ([netcontrol.html](#)) and Starcom ([starcom.html](#))

2.1 Amateur Net Control (<http://192.168.50.1/netcontrol.html>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Create a new net — name, type, frequency, drill flag	
☐	☐	Net appears in the net list and can be selected	
☐	☐	Log a net entry (callsign, name, status, location, remarks)	
☐	☐	Entry appears in the log with UTC timestamp	
☐	☐	Log Traffic button adds a traffic / message entry	
☐	☐	Check-In from roster chip works (pre-populated callsign)	
☐	☐	Remove an entry from the log	
☐	☐	Promote an operator to the roster from a net entry	
☐	☐	Export ICS-309 opens a printable comms log for the selected net	
☐	☐	Export JSON downloads the net data as a file	
☐	☐	Close net marks it inactive	
☐	☐	Observer link (share) opens a read-only observer URL	
☐	☐	Net tabs (Check-Ins / Traffic / All) filter correctly	

2.2 Observer View (<http://192.168.50.1/observer.html>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Page loads without prompting for identity (read-only)	
☐	☐	Active nets list is visible	
☐	☐	Selecting a net shows the live log entries	
☐	☐	Log refreshes automatically without manual reload	
☐	☐	Page loads correctly on a phone-sized screen	

2.3 Starcom Net Logger (<http://192.168.50.1/starcom.html>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Create a Starcom net with dispatch center and Radio IDs	
☐	☐	Log a Starcom entry with Radio ID, unit, status	
☐	☐	Entries show correct UTC timestamps	
☐	☐	Export ICS-309 works for the Starcom net	
☐	☐	Observer link generates a shareable URL	
☐	☐	Close net works correctly	

3 · Resource & Personnel Management

3.1 Member Roster (<http://192.168.50.1/roster.html>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Member list loads from server database	
☐	☐	Add a new member (name, callsign, role, certifications)	
☐	☐	Edit an existing member	
☐	☐	Delete a member	
☐	☐	Check in a member to the active activation	
☐	☐	Walk-in check-in (non-roster member)	
☐	☐	Export CSV downloads the full roster	
☐	☐	Import CSV loads a roster from file	
☐	☐	Export activation log downloads the check-in record	
☐	☐	Filter / search the roster	
☐	☐	Member roles: Net Control (NCS), Operator, Liaison, EC shown correctly	

3.2 Resource Board (<http://192.168.50.1/resources.html>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Resource list loads from server	
☐	☐	Add a resource (personnel / vehicle / equipment / facility)	
☐	☐	Edit a resource	
☐	☐	Delete a resource	
☐	☐	Status pill shows correct color for each state	Available=green, Assigned=blue, Staging=amber, OOS=red, Demob=grey
☐	☐	Click status pill cycles through all 5 states	
☐	☐	Filter by type and status works	
☐	☐	Clear filters restores full list	
☐	☐	Export CSV downloads resource list	
☐	☐	Import CSV loads resources from file	

3.3 Facilities (<http://192.168.50.1/facilities.html>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Seeded facilities load (McHenry County EOC, etc.)	
☐	☐	Add a new facility (name, type, address, lat/lon)	
☐	☐	Edit a facility	
☐	☐	Delete a facility	
☐	☐	Copy address to clipboard	
☐	☐	Export CSV	
☐	☐	Facility detail modal shows all fields	

4 · Maps & Tactical Display

4.1 APRS Tactical Map (<http://192.168.50.1/tactical.html>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Map loads with offline tile layer (USGS Imagery+Topo)	
☐	☐	APRS stations appear as markers from Graywolf	
☐	☐	Clicking a station marker shows callsign and details	
☐	☐	Resource overlays from Resource Board appear on map	
☐	☐	Drawn shapes / overlays can be added	
☐	☐	Clear overlays removes drawn shapes	
☐	☐	Export KML downloads the current overlay set	
☐	☐	Messages tab shows APRS messages	
☐	☐	Settings tab allows tile source and layer changes	
☐	☐	Map pans and zooms smoothly	

4.2 Resource Map / Starcom Map (<http://192.168.50.1/resmap.html>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Map loads with offline tiles	
☐	☐	Add a unit to the map (place a marker)	
☐	☐	Edit a unit's details	
☐	☐	Remove a unit from the map	
☐	☐	Pan to a unit on the map	
☐	☐	Draw a zone polygon	
☐	☐	Zones tab lists drawn zones	
☐	☐	Clear zones removes all drawn zones	
☐	☐	Unit status colors display correctly	

4.3 Grid Square Calculator (<http://192.168.50.1/grid.html>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Convert a lat/lon to Maidenhead grid square	
☐	☐	Convert a grid square to lat/lon	
☐	☐	Calculate distance and bearing between two grid squares	
☐	☐	Use My Location gets device GPS coordinates	Requires browser permission
☐	☐	Result shown on Leaflet map	

5 · ICS Platform

Incident management hub at <http://192.168.50.1/ics/>

5.1 Incident Management ([ics/index.html](http://192.168.50.1/ics/index.html))

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Create a new incident (name, type, location, date)	
☐	☐	Incident appears in the incident list	
☐	☐	Advance operational period increments the period number	
☐	☐	Load existing incident populates all section pages	
☐	☐	Incident list persists after page reload	

5.2 Command Section (ics/command.html)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Incident objectives can be added and deleted	
☐	☐	Safety messages can be added and deleted	
☐	☐	Save persists data for the active incident	
☐	☐	Print produces a clean printable ICS command summary	

5.3 Operations Section (ics/operations.html)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	T-card / resource list loads for the active incident	
☐	☐	Add a resource assignment (callsign, assignment, status)	
☐	☐	Drag-and-drop T-cards between status columns	
☐	☐	Set resource status: available / assigned / staging	
☐	☐	Delete a resource	
☐	☐	Open resource detail modal	

5.4 Planning Section (ics/planning.html)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Resource status tab shows all assigned resources	
☐	☐	IAP tab shows IAP form references and links	
☐	☐	Add a document to the incident document list	
☐	☐	Remove a document from the list	
☐	☐	Save All persists planning data	
☐	☐	ICS form links (ICS-213, 214, 309) open the correct pages	

5.5 Logistics Section (ics/logistics.html)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Communications plan rows can be added and deleted	
☐	☐	Supply requests can be added and deleted	
☐	☐	Meal / feeding records can be added and deleted	
☐	☐	Personnel check-in record can be added	
☐	☐	Data persists after save	

5.6 Finance / Admin Section (ics/finance.html)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Cost tracking entries can be added and deleted	
☐	☐	Time tracking entries can be added and deleted	
☐	☐	Procurement entries can be added and deleted	
☐	☐	Export CSV downloads the finance data	

6 · ICS Forms

6.1 ICS-213 General Message (<http://192.168.50.1/ics213.html>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Form fields all accept input (To, From, positions, date, time, subject, message, reply)	
☐	☐	Auto-fill date/time populates current UTC	
☐	☐	Save to Log saves the form to the local log	
☐	☐	Form log shows saved forms, most recent first	
☐	☐	Load a saved form from the log into the form fields	
☐	☐	Delete a saved form from the log	
☐	☐	Clear Saved clears all saved forms	
☐	☐	Generate Form renders the print-ready ICS-213 layout	
☐	☐	Print button on ICS-213 opens browser print dialog	
☐	☐	Print button on NTS Radiogram opens browser print dialog	
☐	☐	Print Center loads with all document categories visible	
☐	☐	Incident Cover Sheet generates and prints correctly	
☐	☐	If CUPS installed: http://192.168.50.1:631 loads CUPS admin UI	Option B — USB printer
☐	☐	If CUPS installed: USB printer visible in CUPS printer list	Option B
☐	☐	If CUPS installed: test page prints successfully from CUPS UI	Option B
☐	☐	If CUPS installed: Windows device can discover and print to shared printer	Option B
☐	☐	If network printer: printer visible in browser print dialog from EMCOMM-NET device	Option C
☐	☐	Clear Form resets all fields	

6.2 ICS-214 Activity Log (<http://192.168.50.1/ics214.html>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Header fields accept input (incident, unit, leader, op period)	
☐	☐	Add Personnel adds a row (name, position, agency)	
☐	☐	Remove Personnel removes a row	
☐	☐	Add Entry adds an activity log row	
☐	☐	Auto Timestamp populates current UTC on a row	
☐	☐	Remove Entry removes an activity row	
☐	☐	Generate Form renders the print-ready ICS-214	
☐	☐	Print works correctly	
☐	☐	Save Local persists the form on the device	
☐	☐	Clear Form resets everything	

6.3 ICS-309 Communications Log (<http://192.168.50.1/ics309.html>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Header fields accept input (incident, op period, operator, station)	
☐	☐	Add Entry adds a log row (date/time, from, to, subject)	
☐	☐	Add (timestamp now) adds a row with current UTC timestamp	
☐	☐	Remove Entry removes a row	
☐	☐	Generate Form renders the print-ready ICS-309	
☐	☐	Print works correctly	
☐	☐	Save to Incident posts the form to the server	
☐	☐	Back to Edit returns to the editable form	

6.4 Planning P (<http://192.168.50.1/ics/planningp.html>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Page loads with Planning P reference diagram image on the left	
☐	☐	All 15 phase buttons present in the center panel	
☐	☐	Phase buttons organized in 5 color-coded groups (gray/yellow/red/green/teal)	
☐	☐	Clicking a phase highlights it and populates the right detail panel	
☐	☐	Detail panel shows Standard Agenda, Required Forms, and Who Should Attend	
☐	☐	Form chips in detail panel open the correct form page when clicked	
☐	☐	Generate briefing sheet opens printable one-page cover for the selected phase	
☐	☐	Briefing sheet includes phase name, agenda, forms table, and attendance table	

6.5 NTS Radiogram (<http://192.168.50.1/nts.html>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Auto-fill date/time populates current date and time	
☐	☐	Auto-fill number generates a sequence number	
☐	☐	All preamble fields accept input (number, precedence, handling, origin, check, place, time)	
☐	☐	Address and text fields accept input	
☐	☐	Generate Radiogram renders the formatted radiogram	
☐	☐	Save to Log saves the radiogram	
☐	☐	Load from Log restores a saved radiogram	
☐	☐	Delete a saved radiogram	
☐	☐	Copy Text copies the radiogram to clipboard	
☐	☐	Clear Form resets all fields	

7 · Winlink

7.1 Pat Winlink Backup (<http://192.168.50.1:8090>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Pat web interface loads in browser	
☐	☐	Can compose and send a test message via Telnet session	Internet required for Telnet
☐	☐	Inbox shows received messages	
☐	☐	Pat service shown as active in dashboard health monitor	

7.2 Winlink Form Import (<http://192.168.50.1/winlink-import.html>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Page loads from dashboard ICS mode card	
☐	☐	Drop zone accepts an RMS_Express_Form XML file	
☐	☐	Paste text area accepts pasted XML or message text	
☐	☐	Parse Form Data button processes the input	
☐	☐	ICS-213 form is detected and labeled correctly	
☐	☐	ICS-214 form is detected and labeled correctly	
☐	☐	ICS-309 form is detected and labeled correctly	
☐	☐	Unknown form type shows a warning (not an error)	
☐	☐	All mapped fields appear in the editable review grid	
☐	☐	Unmapped fields appear in the "not automatically mapped" box	
☐	☐	Incident dropdown loads active incidents from the server	
☐	☐	Direction (sent/received) can be selected	
☐	☐	Open in ICS form routes ICS-213 to ics213.html with fields pre-filled	
☐	☐	Open in ICS form routes ICS-214 to ics214.html with fields pre-filled	
☐	☐	Open in ICS form routes ICS-309 to ics309.html with rows populated	
☐	☐	Archive to incident saves the form to the server and shows success	
☐	☐	If server unreachable, local fallback copy is saved and a warning shown	
☐	☐	Clear resets the importer to blank state	

8 · Callsign & Radio Tools

8.1 Callsign Lookup (<http://192.168.50.1/callsign.html>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Basic callsign search returns name, address, license class	Uses local FCC DB — no internet
☐	☐	Advanced search (name, city, state, license class) works	
☐	☐	Clear Advanced resets the advanced form	
☐	☐	Copy Info copies the result to clipboard	
☐	☐	Add to Roster adds the callsign to the member roster	
☐	☐	Recent searches appear as quick-access chips	
☐	☐	Direct lookup from a chip works	
☐	☐	Error shown if callsign not found in local DB	

8.2 Repeater Database (<http://192.168.50.1/repeaters.html>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Page loads and shows the three source tabs (Offline File, Server API, Sample)	
☐	☐	Sample data tab shows SAMPLE-1/2/3 placeholders with "not a real repeater" warning	
☐	☐	Offline File tab: drag-and-drop a RepeaterBook CSV loads and displays repeaters	
☐	☐	Filter by band (2m, 70cm, etc.) works	
☐	☐	Filter by state works	
☐	☐	Filter by ARES / SKYWARN / RACES works	
☐	☐	Filter by mode (FM, DMR, C4FM, etc.) works	
☐	☐	Sort by callsign, frequency, and city works	
☐	☐	Sort by distance (nearest first) works	
☐	☐	Clicking a repeater row opens the detail panel	
☐	☐	Copy button copies repeater details to clipboard	
☐	☐	Program button shows programming info	
☐	☐	Export CSV downloads the filtered list	
☐	☐	Server API tab: if database populated, loads repeaters from server	Requires fetch_repeaters.py run

8.3 HF Propagation (<http://192.168.50.1/propagation.html>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Page loads and shows band/path controls	
☐	☐	Fetch All generates propagation predictions	Offline ITU-R model — no internet
☐	☐	24-hour heatmap displays for selected band	
☐	☐	MUF and LUF values shown	
☐	☐	Path can be changed (home station to target)	

9 · Reference, Print & Kiwix

9.1 Reference Library (<http://192.168.50.1/refs.html>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Document list loads from server	
☐	☐	Upload a document (PDF, DOCX, image) via drag-and-drop	
☐	☐	Upload via Browse button works	
☐	☐	PDF thumbnail generated after upload	
☐	☐	Filter by category (radio, ics, agency, training, plans, other)	
☐	☐	Search by keyword finds matching documents	
☐	☐	Clear search restores full list	
☐	☐	View document detail (title, description, tags)	
☐	☐	Edit document metadata	
☐	☐	Download a document	
☐	☐	Delete a document	

9.2 Cheat Sheets (<http://192.168.50.1/cheatsheets.html>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Phonetic alphabet tab displays NATO alphabet	
☐	☐	Q-Codes tab displays common Q-codes	
☐	☐	Prowords tab displays procedure words	
☐	☐	ICS tab displays ICS structure and form reference	
☐	☐	CTCSS tab displays tone table	
☐	☐	Band Plan tab displays frequency reference	
☐	☐	Precedence tab displays NTS precedence guide	
☐	☐	Signal Reports tab displays RST / FM signal report scale	
☐	☐	Print produces a clean single-page printout	

9.3 Print Center (<http://192.168.50.1/printcenter.html>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	ICS-213 preview and print works	
☐	☐	ICS-214 preview and print works	
☐	☐	NTS Radiogram preview and print works	
☐	☐	Pre-Flight Checklist preview and print works	
☐	☐	Phonetic Alphabet pocket card prints	
☐	☐	Q-Codes & Prowords reference prints	
☐	☐	ICS Structure & Forms reference prints	
☐	☐	CTCSS / Signal Reports reference prints	
☐	☐	Net Control Log ICS-309 export and print works	
☐	☐	Starcom Net Log ICS-309 export and print works	
☐	☐	Roster / Member Directory prints	
☐	☐	Resource Board status print works	
☐	☐	Incident cover sheet (Generate Cover) creates a cover page	
☐	☐	Clear Cover resets the cover page fields	

9.4 Kiwix Offline Library (<http://192.168.50.1:8081>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Kiwix interface loads at port 8081	
☐	☐	At least one ZIM library is available (Wikipedia, First Aid, or other)	Depends on tier installed
☐	☐	Search within a ZIM library returns results	
☐	☐	Kiwix service shown as active in health monitor	

10 · Dead Man's Switch (<http://192.168.50.1/deadmans.html>)

P	F	FEATURE / FUNCTION	NOTES
☐	☐	Page loads and shows configured nets	
☐	☐	Arm a net for DMS monitoring	
☐	☐	Disarm a net removes it from monitoring	
☐	☐	Reset a net resets its inactivity timer	
☐	☐	Threshold alert fires after configured inactivity period	Set a short threshold to test
☐	☐	Alert log shows triggered events with timestamps	
☐	☐	Clear Log removes the alert history	
☐	☐	Save Config persists threshold settings	

☐	☐	DMS service shown as active in health monitor	
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11 · Operator Identity

P	F	FEATURE / FUNCTION	NOTES
☐	☐	First load prompts for operator identity (callsign, name, role)	
☐	☐	Identity persists across page reloads and navigation	
☐	☐	Callsign shown in dashboard header after setup	
☐	☐	Role shown correctly: NCS / Operator / Liaison / EC	
☐	☐	Identity can be changed / reset	
☐	☐	Observer page does NOT prompt for identity (read-only)	

12 · System Services

Verify via `systemctl status <service>` on the Pi, or via the dashboard health monitor.

P	F	FEATURE / FUNCTION	NOTES
☐	☐	fcc-lookup.service — main API running on port 5050	
☐	☐	health-monitor.service — health API running on port 5051	
☐	☐	ics-platform.service — ICS platform API running on port 5055	
☐	☐	fieldcomms-refs.service — Reference Library API running on port 5056	
☐	☐	deadmans.service — Dead Man's Switch background monitor	
☐	☐	nginx.service — web server proxying all pages	
☐	☐	kiwix.service — Kiwix offline library on port 8081	
☐	☐	tile_server — offline map tile server on port 8083	
☐	☐	graywolf (Graywolf APRS) — APRS TNC on port 8080	If installed
☐	☐	gpsd.service — GPS daemon (if USB GPS connected)	
☐	☐	fcc-refresh.timer — weekly FCC database rebuild fires on schedule	
☐	☐	repeater-refresh.timer — monthly RepeaterBook fetch fires on schedule	
☐	☐	fieldcomms-backup@.service — automated backup runs correctly	

Beta Test Sign-Off

Section	Items	Pass	Fail	Not Tested
1 · System & Infrastructure				
2 · Net Control Logging				
3 · Resource & Personnel				
4 · Maps & Tactical				
5 · ICS Platform				
6 · ICS Forms				
7 · Winlink				
8 · Callsign & Radio Tools				
9 · Reference, Print & Kiwix				
10 · Dead Man's Switch				
11 · Operator Identity				
12 · System Services				
TOTALS				

Failures requiring follow-up:	
Signed off by (name / callsign):	
Date:	

Submit completed checklist (photo or scan) to the project lead. Include this sheet plus any additional failure notes. Reference the user manual for expected behavior of any item.

Operator Workstation Options — Raspberry Pi 500 / 500+ + Raspberry Pi Monitor

FieldComms is a browser-based system — any device with a modern browser and Wi-Fi can connect to EMCOMM-NET and access every tool. The Raspberry Pi 500 paired with the official Raspberry Pi Monitor provides a purpose-built, low-cost, low-power operator workstation that is fully Pi ecosystem compatible and ideal for fixed EOC positions, shelter check-in stations, and net control operator desks. Funding through ARPA-E grants, FEMA BRIC, or 501(c)(3) equipment donations may cover these workstations.

Pi 500 vs Pi 500+ vs Pi 400 — Comparison

SPEC	Pi 400 (2020)	Pi 500 (Dec 2024) ★ Recommended	Pi 500+ (Sept 2025)
Processor	Cortex-A72 @ 1.8 GHz (Pi 4 generation)	Cortex-A76 @ 2.4 GHz (Pi 5 generation)	Cortex-A76 @ 2.4 GHz (Pi 5 generation)
RAM	4 GB DDR4	8 GB LPDDR4X	16 GB LPDDR4X
Storage	microSD only	microSD only (32 GB A2 included)	256 GB NVMe built-in + microSD slot
Keyboard	Membrane	Membrane	Mechanical (Gateron Blue KS-33)
Wi-Fi	802.11ac dual-band	802.11ac dual-band	802.11ac dual-band
Ethernet	Gigabit	Gigabit	Gigabit
USB ports	2× USB 3.0, 1× USB 2.0	2× USB 3.0, 1× USB 2.0	2× USB 3.0, 1× USB 2.0
Display output	2× micro-HDMI	2× micro-HDMI	2× micro-HDMI
Production until	Jan 2028	TBD (current model)	TBD (current model)
Best for FieldComms?	Acceptable — older chip, only 4 GB RAM	✓ YES — Pi 5 chip, 8 GB RAM, current generation	Overkill for browser use Justified only if also running heavy Linux apps locally

Raspberry Pi Monitor — 15.6"

SPEC	DETAIL
Panel	15.6" IPS LCD, 1920×1080 Full HD @ 60 Hz
Video input	Standard HDMI 1.4 (full-size) — requires micro-HDMI to HDMI cable with Pi 500
Audio	2× front-facing stereo speakers, 3.5mm headphone jack
Power	USB-C, 5V/1.5A — can power from Pi 500 USB port at 60% brightness/volume, or from separate USB-C supply for 100% brightness/volume
Mounting	Integrated angle-adjustable kickstand, VESA mount, wall hanger
Dimensions	Slim 15.6" form — similar footprint to a laptop
Colors	Red/White (matching Pi 500 keyboard) or Black
Where to buy	raspberrypi.com · Adafruit · PiShop.us · GigaParts · Amazon

The Pi 500 has micro-HDMI ports; the Raspberry Pi Monitor has a standard full-size HDMI input. You need a micro-HDMI to standard HDMI cable (or adapter). The Pi 500 desktop kit (\$120) includes one. When ordering separately, confirm the cable is included or purchase one — they are available from the same retailers listed above.

Operator Workstation — Bill of Materials (Per Station)

ITEM	MODEL / SPEC	WHERE TO BUY
— Standard Workstation (Pi 500) —		
Raspberry Pi 500 keyboard computer	Pi 500 standalone — Pi 5 chip, 8 GB RAM, 32 GB A2 microSD, keyboard	raspberrypi.com Adafruit · Amazon
Raspberry Pi Monitor	15.6" Full HD IPS, built-in speakers, kickstand, VESA mount	raspberrypi.com GigaParts · Amazon
micro-HDMI to HDMI cable	1m micro-HDMI to standard HDMI (Pi 500 → Monitor) Included in Pi 500 Kit (\$120) — verify before ordering separately	raspberrypi.com Amazon
USB mouse	Any USB or Bluetooth mouse — included in Pi 500 Kit	Amazon
USB-C power supply for Pi 500	Official Raspberry Pi 27W USB-C PD power supply (or equivalent)	raspberrypi.com Amazon
USB-C power supply for monitor (optional, for full brightness)	5V/1.5A USB-C supply — skip if powering monitor from Pi USB port (Pi USB port gives 60% brightness which may be dim outdoors)	Amazon
— Premium Workstation (Pi 500+) —		
Raspberry Pi 500+ keyboard computer	Pi 500+ — Pi 5 chip, 16 GB RAM, 256 GB NVMe SSD, mechanical keyboard (Gateron Blue), RGB	raspberrypi.com Adafruit · Amazon
Raspberry Pi Monitor + cables + mouse	Same as Standard Workstation above	Same as above

501(c)(3) Grant Funding Note: MCEMV/MCEMA is a 501(c)(3) nonprofit. Operator workstation equipment may be eligible for funding through: FEMA BRIC (Building Resilient Infrastructure & Communities), ARPA-E Emergency Communications grants, ARRL Foundation grants, local Community Foundation equipment grants, or AUXCOMM/ARES regional funding. Document each workstation as "emergency communications operator terminal for the McHenry County Emergency Services Volunteers (K9ESV), a 501(c)(3) nonprofit supporting McHenry County Emergency Management Agency (MCEMA) RACES/ARES/Starcom communications."

Recommended Deployment by Position

POSITION	RECOMMENDED DEVICE	REASON
Net Control Operator (fixed EOC desk)	Pi 500 + Pi Monitor	Dedicated keyboard, good display, Chromium runs every FieldComms page smoothly
ICS Section Chief (Planning, Ops, Logistics)	Pi 500 + Pi Monitor or Windows/Mac laptop	Needs reliable keyboard for ICS form entry; Pi 500 is sufficient for all ICS platform pages
Served Agency Staff (EOC or shelter)	Any tablet, Chromebook, or Pi 500	Observer Mode and ICS forms work on any browser — no special hardware needed
Winlink / JS8Call Operator	Windows laptop (required)	Winlink Express and JS8Call are Windows applications — Pi 500 cannot run them natively

Mobile / Field Operator	Phone or tablet	FieldComms works on any smartphone browser on EMCOMM-NET — no dedicated hardware needed
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■ The Pi 500 cannot run Windows applications natively. Winlink Express and JS8Call must run on the Windows laptop connected to the IC-7300. The Pi 500 is ideal for all browser-based FieldComms tools — net control logging, ICS forms, tactical maps, resource tracking, and the full ICS platform.